Society of Petroleum Engineers

SPE-227235-MS

Novel Real-Time Data Analytics Approach for Optimizing Drilling Operations: A Case Study

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

The drilling rig is equipped with a huge number of data sensors that record the operation`s functionalities of different rig systems, personnel, and equipment. Integrating such data, transmitting through data transformation protocols, and accomplishing real-time data analytics will greatly contribute to enhancing the drilling operation performance, mitigating the drilling hazards, minimizing invisible lost time (ILT), and reducing the operation cost. Real-time data analytics show the key performance indicators (KPIs) for every operation on the rig in terms of drilling, tripping, connection, etc., to reveal the data insights for the best decision-making using the advanced visualization dashboards and reporting system. The current research shows the high capabilities of data analytics in real-time through a novel integrated approach for multi-source data toward enhancing the operation performance and reducing the drilling cost with a real field case study.

This paper presents an integrated systematic data analytics workflow for real-time rig sensors and daily drilling reports to efficiently monitor and enhance operations performance. The novel developed approach provides an accurate measurement of operations indicators and identification of potential improvements for the drilling operation. The system aggregates and processes the data transmitted from the drilling rigs and integrates it with daily drilling reporting. Operation performance is displayed through benchmark analyses from all rigs, by operation type, and equipment, giving a statistical indication of the technical limit to reduce ILT. The paper shows a real field case study to present drilling operation efficiency enhancement through real-time data analytics.

25% of the drilling operation is considered ILT, and this might be reduced through advanced data analytics for the drilling data. The developed approach provides impartial and detailed KPIs on operation performance to determine and eliminate ILT. Through a case study, the results showed an enhancement of up to 10% in time-saving and cost reduction. This amount of time is saved by adjusting these parameters in real-time based on the insights provided by the analytics system, resulting in optimized efficiency and cost reduction. The developed approach is currently upgraded with developed machine-learning models for prediction and classification purposes through advanced computational algorithms to predict the rate of penetration and drilling hazards and auto-detect the drilled formation type/class from collected drilling data.

Overall, the novel approach presented within the study shows that real-time data analytics can significantly improve drilling efficiency by identifying patterns and trends in data and providing actionable insights that can be used to optimize drilling operations. This has the potential to reduce drilling costs, improve safety, and increase the overall efficiency of drilling operations.

Key Words: Drilling KPIs, Operation Performance, Real-Time, Data Analytics, Case Study, Operational Efficiency, Cost Optimization

Introduction

The modern drilling rig is a complex and sophisticated system equipped with a vast array of sensors that continuously record operational parameters and functionalities across various rig systems, personnel activities, and equipment performance (Zausa et al., 2018; Gamal et al., 2023). This proliferation of data presents a significant opportunity to move beyond traditional, often reactive, methods of drilling management towards a more proactive and data-driven approach. The ability to effectively integrate such diverse data streams, transmit them through efficient data transformation protocols, and perform real-time data analytics holds immense potential for fundamentally enhancing drilling operation performance, proactively mitigating drilling hazards, minimizing the often-overlooked phenomenon of invisible lost time (ILT), and ultimately achieving substantial reductions in overall operation cost (Gamal et al., 2024).

Traditional drilling performance analysis often relies on post-job reviews of daily drilling reports and summary statistics, which can mask subtle inefficiencies and do not provide the real-time insights necessary for dynamic optimization. Real-time data analytics, in contrast, enables the continuous monitoring and analysis of operational data as it is being generated. This allows for the calculation and visualization of key performance indicators (KPIs) for every specific operation on the rig, encompassing drilling, tripping in and out, connection-making, and other critical activities (Svensson et al., 2015). By providing advanced visualization dashboards and reporting systems, real-time analytics can reveal crucial data insights into ongoing performance, facilitating faster and more informed decision-making by the operational team and onshore support personnel.

Invisible lost time (ILT) is estimated to account for a significant portion, potentially up to 25%, of the total operational time in drilling. This subtle yet substantial inefficiency can be significantly reduced through the application of advanced data analytics to the vast amounts of drilling data being collected. The developed approach provides impartial, detailed, and continuously updated KPIs on operational performance. By quantifying the time spent on various micro-activities within each operation type, the system can determine and highlight instances of ILT, enabling operators to take targeted action to eliminate these inefficiencies(Basbar et al., 2016; Ferrari et al., 2022).

Invisible Lost Time

Invisible lost time (ILT) is technically defined as the measurable discrepancy between the theoretical minimum time required to execute a specific drilling operation under optimal conditions and the actual time consumed, explicitly excluding non-productive time (NPT) events. Unlike NPT, which encompasses major, identifiable disruptions such as equipment failures or well-control incidents, ILT arises from subtle inefficiencies embedded within routine operations. Key characteristics of ILT include its inherent subtlety, where events manifest as brief, often unreported delays (e.g., 30-second lags during pipe handling) that evade detection in traditional daily drilling reports (DDRs). Its cumulative nature magnifies its impact: though individual instances appear negligible, ILT compounds significantly over hundreds of repetitions, such as connections in a deep well, potentially wasting 25% of operational time. Additionally, ILT exhibits high variability, fluctuating with rig crew expertise, equipment reliability, wellbore conditions, and supervisory decisions, rendering it context-dependent and non-uniform. Crucially, ILT remains elusive without high-resolution data analytics, as traditional monitoring methods lack the granularity to detect it. Common examples include inefficient drill-pipe handling during connections, unnecessarily slow tripping speeds not justified by wellbore stability, prolonged fluid conditioning beyond technical requirements, conservative drilling parameters that reduce the rate of penetration (ROP), communication delays between crews, and suboptimal hole-cleaning practices during circulation.

Quantifying ILT: A Data-Driven Methodology

Accurate quantification of ILT demands a rigorous, data-centric methodology that transcends conventional time-tracking approaches. The process begins with detailed operational segmentation, breaking down macro-operations like "tripping in" or "making a connection" into granular micro-activities such as "hoisting drill string," "setting slips," or "breaking a connection". This decomposition enables precise measurement through high-frequency data acquisition from real-time sensors, including hookload, block position, pump pressure, torque, and RPM, which timestamp micro-activity durations at 1–10-second intervals. Establishing technical limits or benchmarks for each micro-activity is foundational, derived from engineering simulations (e.g., hydraulic models for safe tripping speeds), historical top-quartile performance data from analogous operations, or manufacturer specifications for rig capabilities. ILT is then calculated as the sum of deviations between actual performance and these benchmarks across all micro-activities, rigorously excluding periods classified as NPT. For instance, if the technical limit for a connection is 2.5 minutes (based on best-performing crews) but actual performance averages 3.1 minutes, the 0.6-minute delta per connection represents quantifiable ILT. This methodology transforms ILT from an abstract concept into a precise metric, enabling targeted interventions to eliminate micro-inefficiencies systematically.

ILT Impact on Well Delivery Performance

The pervasive presence of ILT exerts a direct and profoundly negative impact on the overall efficiency, cost, and schedule of well delivery. Its cumulative nature translates into significant tangible consequences: Primarily, ILT substantially increases operational time, adding unnecessary hours or even days to the drilling schedule and extending the total duration of the well construction project. This extended time-on-well directly increases rig costs, as rig rates are typically charged on a daily or hourly basis, leading to a linear escalation in total expenditure solely attributable to inefficiency. Furthermore, the extended well delivery schedule inevitably delays the commencement of critical operations, whether it be hydrocarbon production that directly impacts project revenue and net present value (NPV), or the start of essential water injection for reservoir management, thereby affecting broader field development economics and return on investment (ROI).

The longer the rig is operational due to ILT, the greater the exposure to potential drilling hazards (such as stuck pipe, kicks, or wellbore instability) and equipment failures, inherently increasing operational risk and the potential for more severe NPT. Fundamentally, ILT represents a reduction in overall operational efficiency, signifying the suboptimal utilization of highly valuable rig time, skilled personnel, and associated equipment. Consequently, systematically identifying and eliminating ILT is not merely an efficiency initiative; it is a critical pathway to achieving genuine operational excellence and realizing substantial, measurable cost reductions throughout the drilling lifecycle.

Key Performance Indicators (KPIs) for Drilling Optimization and ILT Reduction

Key performance indicators (KPIs) serve as quantifiable metrics essential for evaluating the effectiveness of specific activities and processes within drilling operations. Robust, real-time KPIs are fundamental for monitoring operational progress, identifying deviations from optimal performance, and uncovering opportunities for improvement, particularly in reducing invisible lost time. The implementation of a real-time data analytics approach generates a comprehensive suite of KPIs, delivering objective insights into operational efficiency and enabling data-driven decision-making. Drilling KPIs can be systematically classified based on either the operational phase or the performance aspect they measure as shown in Figure 1.

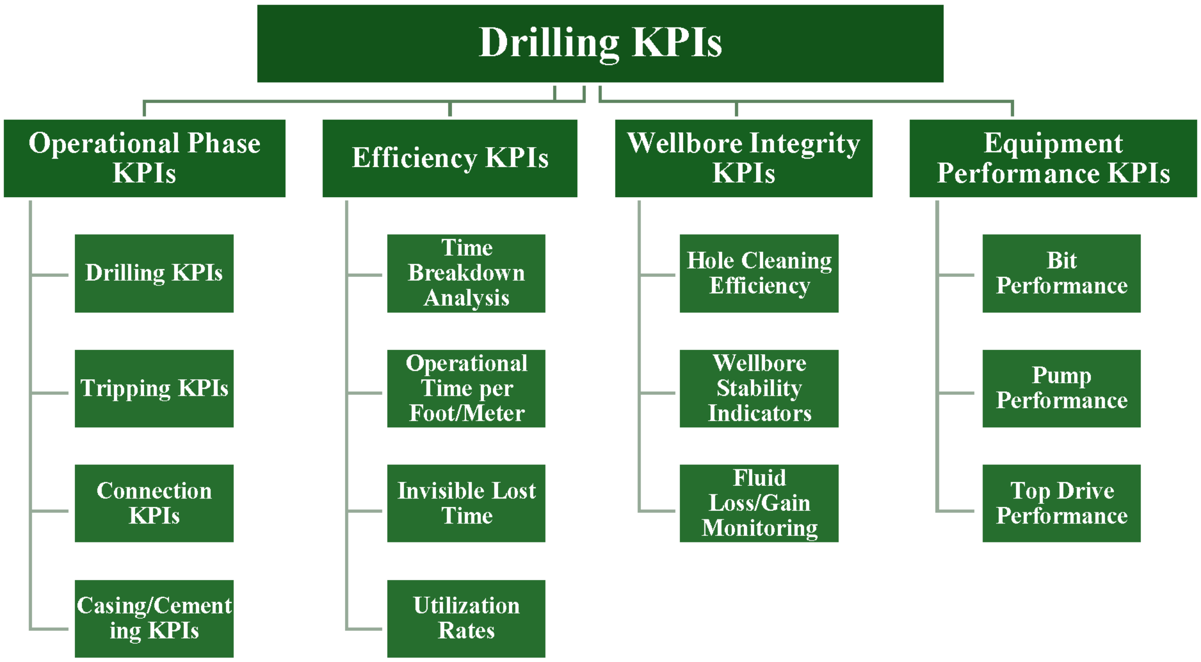

Figure 1—Drilling KPIs

Operational Phase KPIs

Drilling KPIs can be classified into four primary categories based on operational phases or performance aspects. Operational phase KPIs encompass metrics tied to specific drilling activities. Drilling operations are evaluated through rate of penetration (ROP in terms of instantaneous and average), rotary speed (RPM), weight on bit (WOB), surface/downhole torque, standpipe pressure (SPP), flow rate, mechanical specific energy (MSE), bit hydraulics, and Vibration levels (axial, lateral, torsional). Tripping operations utilize tripping speed (in/out), time per stand, time per trip, and hookload analysis to assess friction and sticking risks. Connection operations rely on time per connection, often dissected into micro-activities like slip-to-slip time and make-up time. The casing and cementing phases employ casing running speed, centralization effectiveness, cement placement time, and displacement efficiency to ensure technical compliance.

Efficiency KPIs

Efficiency KPIs evaluate resource utilization and time management, including operational time per foot/meter to measure drilling efficiency, time breakdown analysis to quantify activity distribution (e.g., drilling vs. tripping), invisible lost time (ILT) per operation type to isolate cumulative inefficiencies, and utilization rates for rig equipment and personnel productivity. Wellbore integrity KPIs monitor safety-critical conditions such as hole cleaning efficiency (via cuttings returns or fill-on-bottom measurements), wellbore stability indicators (tracking torque/drag trends, shale shaker performance, and connection gas), and fluid loss/gain monitoring for early influx or loss detection. Equipment Performance KPIs focus on asset reliability: bit performance is gauged by footage drilled per run and ROP trends; pump performance tracks volumetric efficiency and runtime, while top drive performance assesses operational efficiency and runtime. This structured KPI framework forms the analytical backbone for identifying ILT, benchmarking against technical limits, and driving real-time operational improvements.

Wellbore Integrity KPIs

Wellbore Integrity KPIs provide critical insights into the stability and safety of the wellbore during drilling operations. Hole cleaning efficiency is monitored through cuttings return analysis (quantifying the volume of drilled solids reaching the surface relative to theoretical yield) and fill-on-bottom measurements (detecting sediment accumulation after tripping operations). These metrics directly indicate whether annular velocities and fluid properties are sufficient to prevent pack-offs or stuck pipe incidents. Wellbore Stability Indicators encompass torque and drag trends (deviations from expected friction factors signaling borehole narrowing or collapse), shale shaker performance (evaluating solids removal efficiency via cuttings volume and size distribution), and connection gas detection (identifying hydrocarbon influx during pump-off periods, revealing formation pressure imbalances). Fluid loss/gain monitoring tracks real-time pit volume discrepancies to provide early warnings of lost circulation or kicks, enabling immediate well-control responses.

Equipment Performance KPIs

Equipment Performance KPIs focus on the reliability and efficiency of critical drilling assets. Bit performance is assessed through footage drilled per run (total meters drilled before replacement) and ROP trends (tracking penetration rate consistency or decline to identify wear, balling, or suboptimal parameter selection). Pump Performance combines volumetric efficiency (calculated output versus theoretical displacement) with runtime availability (percentage of operational time pumps function without failure) to ensure hydraulic requirements are met while minimizing maintenance downtime. Top drive performance utilizes operational efficiency (ratio of productive rotation/hoisting time to total scheduled runtime) and runtime reliability (tracking unscheduled repairs or stalls) to safeguard continuous drilling and tripping operations.

Systematic Data Analytics Workflow

The current research presented in this paper demonstrates the high capabilities of real-time data analytics through a novel integrated approach designed for the efficient aggregation and analysis of multi-source data streams originating from the drilling rig. This approach is specifically aimed at enhancing operational performance and achieving significant reductions in drilling costs. The core of the novel approach is a systematic data analytics workflow (Figure 2) designed to efficiently process multi-source data in real-time to support operational decision-making and performance enhancement. This workflow encompasses data acquisition, processing, analysis, and visualization.

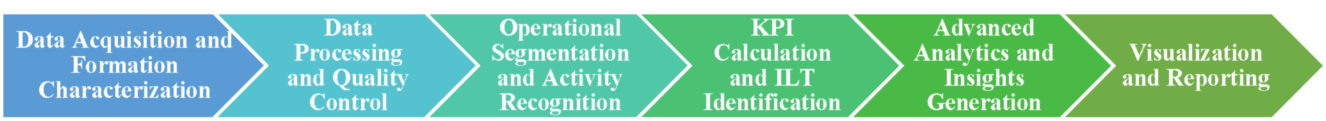

Figure 2—Technical Workflow and Engineering Methodology

Data Acquisition and Integration

The workflow initiates with the robust acquisition of data from multiple rig sources. This includes capturing high-frequency data streams (typically 1 Hz to 100 Hz or higher) from real-time sensors monitoring the drilling control system (DCS), mud system, standpipe, pumps, and potentially downhole tools (MWD/LWD), providing granular operational visibility. Structured data from daily drilling reports (DDRs), encompassing time breakdowns, reported non-productive time (NPT) events, operational remarks, and formation tops, is also integrated; this integration allows for validation of sensor data and provides essential context for analyzing sensor readings within specific operational phases. Furthermore, other relevant data sources such as well plans (trajectory, casing program, mud program), geological prognoses, historical well data, and equipment maintenance records are incorporated to enrich the analysis. Data from these disparate sources is aggregated and transmitted to a central processing and storage system, frequently leveraging cloud-based infrastructure for scalability and accessibility, with data transformation protocols applied to ensure consistency and compatibility.

Data Processing and Quality Control

Following the acquisition, the raw data undergoes rigorous processing and quality control. This involves data cleaning to identify and handle missing points, outliers, and sensor errors, employing algorithms to interpolate missing values or flag unreliable data. Data filtering and smoothing are applied to remove noise from high-frequency sensor streams while preserving essential operational signals. Crucially, data alignment and synchronization ensure all streams from different sensors and sources are accurately correlated based on precise timestamps. Data validation is performed by cross-referencing information from different sources, such as comparing reported NPT in DDRs with deviations detected in sensor data, to confirm the accuracy of both datasets.

Operational Segmentation and Activity Recognition

A critical step segments the continuous data stream into distinct operational activities (e.g., drilling, tripping in, making connections, reaming). This is achieved through algorithms recognizing patterns in key sensor data, utilizing both rule-based segmentation (defining rules based on changes in specific parameters like hookload, RPM, or flow rate to indicate transitions) and machine learning-based segmentation (employing models trained on historical data to automatically recognize complex patterns indicative of different activities). Precise segmentation is fundamental for accurately calculating Key Performance Indicators (KPIs) and quantifying invisible lost time (ILT) for each specific task.

KPI Calculation and ILT Identification

With processed and segmented data, the workflow calculates real-time KPIs and identifies ILT. Algorithms continuously compute KPIs based on processed sensor data for the current activity (e.g., instantaneous ROP from bit depth changes, connection time duration). These real-time KPIs are benchmarked against established technical limits or historical performance; deviations exceeding defined thresholds are flagged as potential ILT instances. The duration of each deviation during micro-activities is then aggregated to quantify the total ILT for a specific operation or defined period.

Advanced Analytics and Insights Generation

Hydraulic modeling and simulation of the flow behavior of the drilling fluid in the annulus during tripping and circulation to predict equivalent circulating density and optimize pump rates and fluid properties for efficient hole cleaning and pressure management (Alharbi et al., 2025). Figure 3 illustrates hookload simulations and actual measurements during drilling operations for analyzing drag and friction. Multiple lines represent simulated hookloads under various friction factors (FF), as the solid black line represents the Hookload as a reference, while scattered points show the actual measured hookload. Deviations could indicate changes in wellbore conditions, fluid properties, or sticking issues.

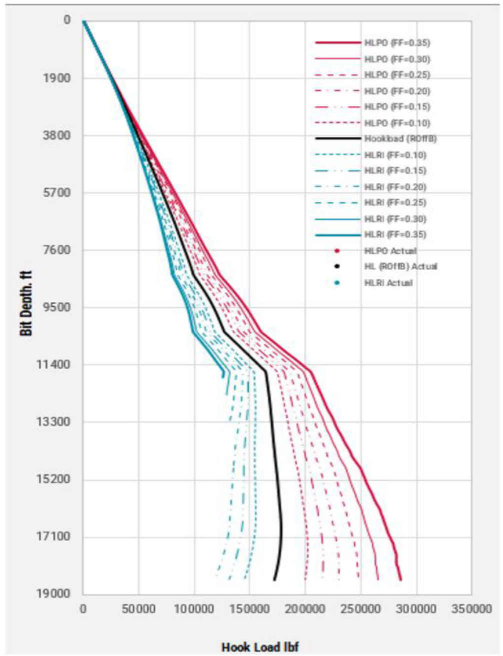

Figure 3—Simulated Hookloads Under Various Friction Factors

A digital platform serves for data management and real-time monitoring throughout the workflow. Data acquisition and aggregation system to capture high-frequency real-time data from all relevant surface sensors (hookload, torque, RPM, pump rate, standpipe pressure, flow in/out) and downhole sensors if available, and streaming the data into a unified database within the digital platform with advanced technological capabilities for optimizing the operation performance(AlBar et al., 2018; Gamal, 2025). Creating intuitive dashboards within the digital platform to provide real-time visibility of key performance indicators (KPIs), fluid properties, and operational parameters for the engineering and operations teams(Ferrari et al., 2021). Figure 4 represents a synthetic sample from the real-time drilling digital platform, providing real-time operational data and performance analysis.

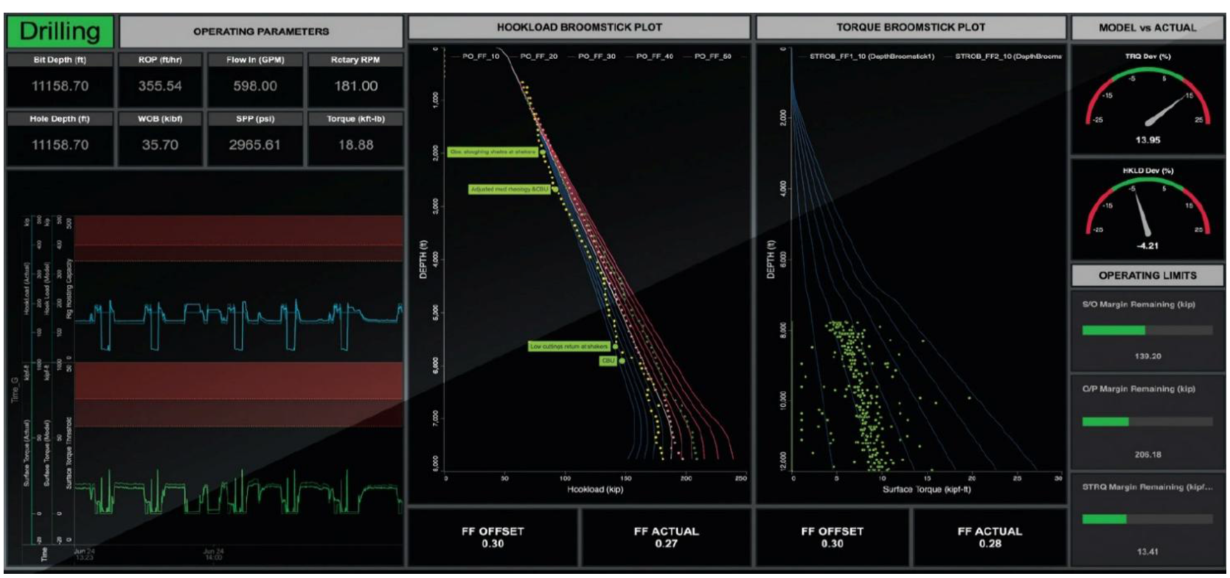

Figure 4—Drilling Digital Platform (Whitfield, 2022)

Beyond basic KPI calculation, the workflow incorporates advanced analytics. Automated root cause analysis algorithms examine sensor data patterns and parameter correlations to identify potential reasons for observed ILT or performance deviations (e.g., correlating high torque with low ROP, linking vibration patterns to reduced efficiency). Predictive modeling, particularly using machine learning, is employed to forecast ROP based on current drilling parameters, formation properties, and bit wear, predict drilling hazards (like stick-slip, wellbore breathing, or connection gas kicks) by identifying precursor patterns, and automatically classify drilled formations based on drilling parameters and subsurface characteristics. Benchmarking analysis provides detailed statistical performance comparisons across wells, rigs, or crews to identify best practices and improvement areas (Cao et al., 2018; de Wardt et al., 2016).

Visualization and Reporting

The final step presents analyzed data and insights through intuitive dashboards and reporting systems. Real-time dashboards provide operational personnel and onshore support with immediate visual feedback on current performance, KPIs, ILT indicators, and potential hazard warnings. Automated performance reports summarize metrics, identify ILT, and provide insights over various periods (e.g., per stand, connection, trip, day, or well section). An alerting system automatically notifies personnel of significant parameter deviations or predicted hazards.

This systematic workflow, spanning data acquisition to visualization and advanced analytics, forms the technical foundation for transforming raw rig data into actionable intelligence aimed at optimizing drilling operations and minimizing invisible lost time.

Case Study Results and Observations

The implementation of the engineered data-centric workflow and integrated planning methodology yielded highly successful operational outcomes, demonstrating tangible improvements in drilling performance and efficiency.

Tripping performance optimization analysis of real-time data from the integrated system revealed significant improvements in tripping efficiency. As illustrated in Figure 5, a substantial reduction in invisible lost time (ILT) associated with tripping operations was achieved. This was evidenced by an increase in average tripping speed from 1646 ft/hr in Q1 2023 to 2098 ft/hr in Q2 2025, representing a marked improvement in operational speed.

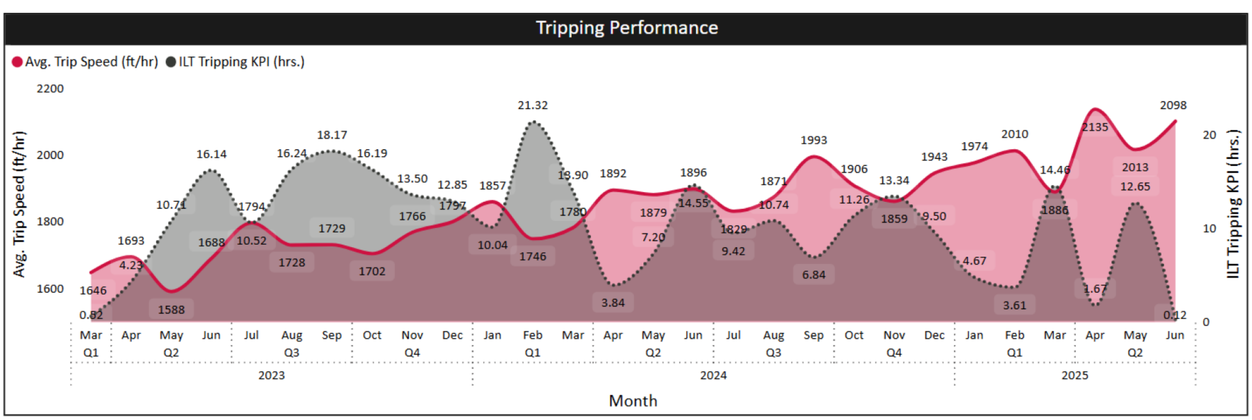

Figure 5—ILT Tripping KPI

Concurrently, Figure 6 highlights the enhanced efficiency of tripping connections. The average tripping connection time was successfully reduced from 2.5 minutes to 1.9 minutes. This quantifiable reduction per connection, accumulated over numerous connections during a well, translates directly into a significant decrease in overall rig operating hours and associated costs.

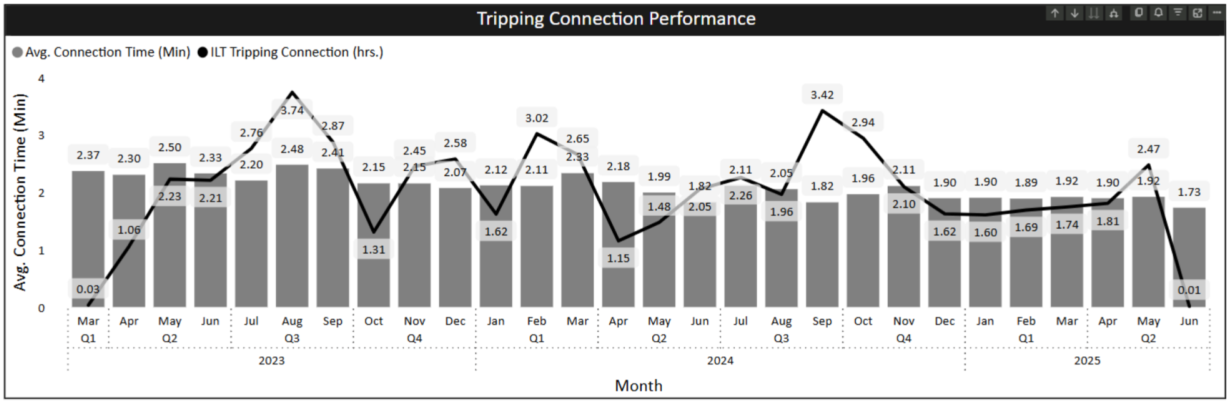

Figure 6—Tripping Connection Performance

The real-time data analytics approach provided unprecedented visibility into flat Key Performance Indicator (KPI) contributors, which represent non-drilling and non-tripping operational inefficiencies. As detailed in Figure 7, the system effectively identified the top 10 flat ILT contributors, pinpointing specific operational segments for improvement. Notably, the top three major flat KPI contributors were identified as pressure test BOP operations, make-up lateral section BHA, and pressure test casing/liner operations. This granular identification of time inefficiencies provides a clear pathway for projecting quantifiable improvements through the revision of current operational procedures and the implementation of optimized, industry-standard practices.

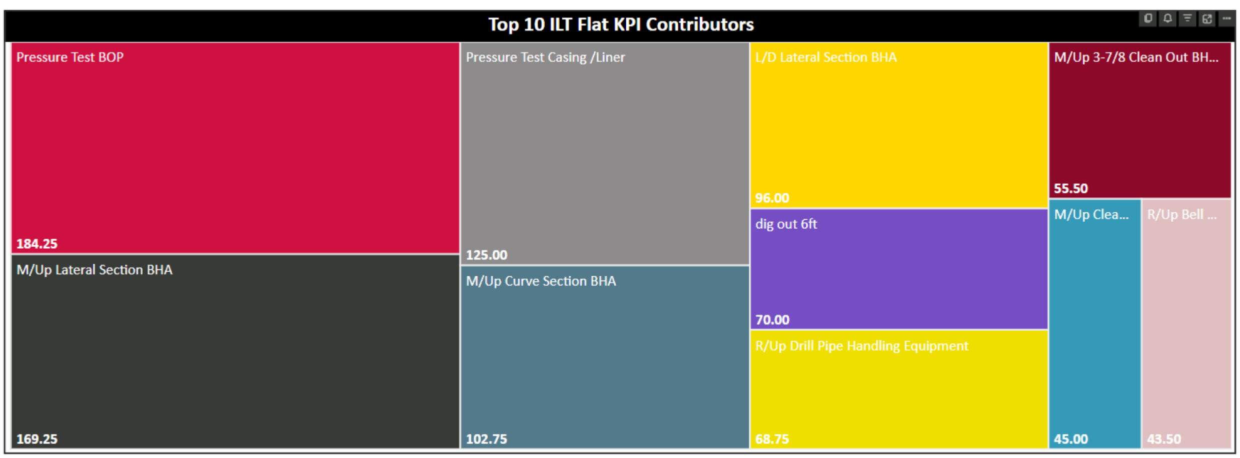

Figure 7—Top 10 ILT Flat KPI Contributors

Conclusions

- The novel real-time data analytics approach presented in this study offers a transformative solution for optimizing drilling operations by systematically identifying and minimizing invisible lost time (ILT) and enhancing overall operational efficiency. By integrating high-frequency data from rig sensors with daily drilling reports and employing a systematic data analytics workflow, the developed system provides unparalleled real-time visibility into operational performance through detailed key performance indicators (KPIs).

- The case study presented provides compelling field-proven evidence of the practical benefits of this approach. By enabling real-time monitoring of KPIs, the system successfully highlighted opportunities for reducing ILT during various drilling operations. The resulting operational adjustments, made dynamically based on the insights provided by the analytics system, led to a quantifiable enhancement in operational efficiency, achieving a time-saving and corresponding cost reduction of up to 10% compared to historical performance in the field.
- This study proves that moving beyond traditional, summary-based performance analysis towards real-time, data-driven insights is crucial for unlocking significant efficiency gains in drilling. The developed approach provides objective and detailed KPIs that make ILT visible and measurable, enabling operators to target specific operational procedures for improvement.
- Furthermore, the ongoing development of machine-learning models for prediction and classification represents a significant step towards enabling proactive hazard mitigation and automated optimization recommendations, further enhancing the value proposition of the system.
- In conclusion, the novel real-time data analytics approach demonstrates significant potential to fundamentally improve drilling efficiency by transforming raw data into actionable intelligence. This has direct implications for reducing drilling costs, improving safety by proactively identifying hazardous conditions, and increasing the overall efficiency and predictability of drilling operations, contributing to more successful and economically viable well delivery. The methodologies presented are scalable and applicable across various drilling environments, offering a pathway to operational excellence in the modern drilling industry.

Acknowledgment

The authors express sincere appreciation to Weatherford Company for its invaluable support in conducting this study and for granting publication permission. The authors also recognize the instrumental contributions of the engineering and operations teams, whose exceptional collaboration and teamwork were fundamental to the successful outcome of this research.

References

1. AlBar, A.H., Alotaibi, B.M., Asfoor, H.M., Nefai, M.S., 2018. A Journey Towards Building Real-Time Big Data Analytics Environment for Drilling Operations: Challenges and Lessons Learned. Presented at the SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition, OnePetro. https://doi.org/10.2118/192285-MS
2. Alharbi, A., Petre, R., Gamal, H., 2025. Casing Exit Excellence: a Holistic Strategy of Integrated Engineering Models and Well Integrity Analytics for Extended Reach Wells.
3. Basbar, A.E., Al Kharusi, A., Al Kindi, A., 2016. Reducing NPT of Rigs Operation through Competency Improvement: A Lean Manufacturing Approach.
4. Cao, D., Loesel, C., Paranji, S., 2018. Rapid Development of Real-Time Drilling Analytics System. Presented at the IADC/SPE Drilling Conference and Exhibition, OnePetro. https://doi.org/10.2118/189595-MS
5. de Wardt, J.P., Rushmore, P.H., Scott, P.W., 2016. True Lies: Measuring Drilling and Completion Efficiency. Presented at the IADC/SPE Drilling Conference and Exhibition, OnePetro. https://doi.org/10.2118/178850-MS
6. Ferrari, F., Fenaroli, D., Michelez, J., Magni, J., 2022. Integrated Rig Management Platform.
7. Ferrari, F., Naselli, R., Brunetti, P., Michelez, J., Zini, E., 2021. Digitalization for Reducing Carbon Footprint in Drilling Operations. Presented at the Abu Dhabi International Petroleum Exhibition & Conference, OnePetro. https://doi.org/10.2118/207407-MS

8. Gamal, H., 2025. Technological Integration and Innovative Strategies Harnessing Artificial Intelligence for Operational Excellence [WWW Document]. https://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/979-8-3693-6447-5.ch009. URL https://www.igi-global.com/gateway/chapter/www.igi-global.com/gateway/chapter/359302 (accessed 5.29.25).
9. Gamal, H., Albahri, S., Essa, H., Srivastava, M., Alghamdi, N., Shaikh, W., 2024. Driving Operational Excellence: Cost Savings and Performance Optimization Through Tripping and Connection Time Optimization.
10. Gamal, H., Elkatatny, S., Al Gharbi, S., 2023. Rig Sensor Data for AI-ML Technology-Based Solutions: Research, Development, and Innovations. Presented at the ADIPEC, OnePetro. https://doi.org/10.2118/216429-MS
11. Svensson, I., Saeverhagen, E., Bouillouta, F.M., 2015. Driving Rig Performance through Real-Time Data Analysis, Benchmarking, Dashboards and Developed Key Performance Indicators. Presented at the SPE Digital Energy Conference and Exhibition, OnePetro.
12. Whitfield, S., 2022. The art of data science: Industry hones in on reliability, ease of use. *Drilling Contractor*. URL https://drillingcontractor.org/the-art-of-data-science-industry-hones-in-on-reliability-ease-of-use-63276 (accessed 6.11.25).
13. Zausa, F., Pellicciotta, L., Spelta, A., Bruni, V., Baio, F., Medea, A., Rossi, N., 2018. Multi Source Data Analysis To Improve Drilling Efficiency, Reduce Costs and Enhance Performance Monitoring. Presented at the Abu Dhabi International Petroleum Exhibition & Conference, OnePetro. https://doi.org/10.2118/192624-MS